

Forecasting USD/VND Exchange Rate Volatility with Global and Domestic Financial Shocks: Evidence from GARCH-Family and EGARCH-X Models

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Abstract—This study evaluates one-step-ahead forecasts of USD/VND exchange-rate volatility using daily observations from January 2010 to December 2025. The empirical design follows a strict chronological train-validation-test framework to prevent future information from affecting model selection. An ARMA(4, 5) conditional-mean specification is selected on the training sample after applying residual-whiteness requirements. Its recursive innovations are then used to estimate and compare Student- t GARCH-family models, directly comparable custom EGARCH and EGARCH-X specifications, and simple variance benchmarks. Model selection is based exclusively on validation-period QLIKE loss. The GJR-GARCH(1, 1)- t model achieves the lowest validation QLIKE and is therefore locked before the untouched 2024–2025 test period is evaluated. After refitting on the combined training and validation sample, the locked model remains competitive and produces lower test QLIKE than the EWMA, rolling-variance, and historical-variance benchmarks. However, reporting-only EGARCH-type comparators obtain numerically lower test losses, indicating that relative forecast performance varies across evaluation periods. The absolute-shock EGARCH-X specification ranks second during validation and first in the squared-return robustness exercise, suggesting that shock magnitudes can contain useful information under selected specifications and volatility proxies. Supplementary dynamic HAC regressions show that volatility persistence explains most of the model-implied volatility path, while the absolute magnitude of lagged VN-Index returns is positively associated with subsequent volatility. Weekly recursively identified VAR results indicate that U.S. Treasury-yield changes, DXY returns, and VN-Index returns predict weekly USD/VND returns, whereas none of the examined shocks significantly predicts weekly conditional volatility. Overall, USD/VND volatility is primarily driven by internal persistence, while global and domestic shocks provide limited and specification-dependent incremental information.

Index Terms—Time Series, USD/VND; exchange rate volatility; GARCH-type models; global shocks; Vietnam; financial risk.

I. INTRODUCTION

Exchange-rate volatility is an important source of macro-financial risk in developing economies. Fluctuations in the exchange rate can affect inflation, trade competitiveness, external-debt valuation, capital flows, and overall financial stability. These effects are particularly relevant in economies where firms and investors have limited access to hedging instruments and domestic financial markets may be less able to absorb external disturbances. Reliable volatility forecasts are therefore important for risk management, financial planning, and policy assessment.

Vietnam provides a relevant setting for studying exchange-rate volatility because the USD/VND exchange rate operates

under a managed exchange-rate environment rather than a fully floating regime. Under such a system, exchange-rate movements may remain limited during normal periods, while larger adjustments can occur during episodes of policy pressure, global financial stress, or changes in international dollar conditions. USD/VND volatility may therefore reflect not only its own persistence but also delayed responses to external and domestic financial shocks.

This study forecasts USD/VND exchange-rate volatility using daily data from January 2010 to December 2025. The dependent series is the daily log return of the USD/VND nominal exchange rate. The forecasting framework incorporates four global financial variables and one domestic financial variable. Changes in the CBOE Volatility Index (VIX) capture global risk aversion. Changes in the U.S. 10-year Treasury yield represent shifts in international interest-rate conditions. Returns on the U.S. Dollar Index (DXY) measure broad movements in U.S. dollar strength. Changes in WTI crude oil prices capture oil-market disturbances. VN-Index returns are included as a proxy for domestic equity-market conditions. These financial shocks are lagged before entering the forecasting models so that each predictor is observable prior to the forecast date.

The empirical design is structured as an out-of-sample volatility-forecasting exercise. The dataset is divided chronologically into training, validation, and test periods. An ARMA conditional-mean model is first estimated to filter predictable short-run dependence in USD/VND returns. The resulting innovations are then used to estimate a set of GARCH-family volatility models, including GARCH, GJR-GARCH, and symmetric and asymmetric EGARCH specifications under alternative innovation distributions. These parametric models are compared with non-parametric variance benchmarks using recursive one-step-ahead forecasts.

The analysis then extends the EGARCH framework by incorporating lagged global and domestic financial shocks directly into the conditional-variance equation. The custom EGARCH-X specifications are designed to test whether financial shocks contain incremental predictive information beyond the exchange rate's own volatility history. Signed-shock models examine whether the direction of financial disturbances matters, while absolute-shock models examine whether shock magnitude matters regardless of direction. Event-control specifications are estimated separately to account for predefined crisis periods and major exchange-rate adjustment episodes.

Model selection is based exclusively on validation-period forecasting performance. The primary ranking criterion is QLIKE loss, which is appropriate for evaluating latent-volatility forecasts using noisy variance proxies. After the preferred specification is locked, the required models are re-estimated using the combined training and validation sample and evaluated once on the untouched test period. This procedure prevents test-period information from influencing model choice and provides a transparent assessment of genuine out-of-sample forecasting performance.

This paper addresses four research questions. First, do USD/VND returns exhibit the statistical properties that justify conditional-volatility modelling, including volatility clustering, conditional heteroskedasticity, and fat-tailed innovations? Second, which GARCH-family specification provides the strongest out-of-sample volatility forecasts after conditional-mean filtering? Third, do lagged global and domestic financial shocks improve USD/VND volatility forecasts relative to a comparable no- X EGARCH benchmark? Fourth, are absolute shock magnitudes more informative than signed shock directions for forecasting future volatility?

The study contributes to the literature in three ways. First, it evaluates USD/VND volatility within a strict train-validation-test framework, shifting the focus from in-sample model fit to out-of-sample forecast performance. Second, it compares standard GARCH-family models with directly comparable custom EGARCH- X specifications, allowing the incremental predictive value of global and domestic financial shocks to be assessed explicitly. Third, it distinguishes between signed and absolute shocks, providing evidence on whether the direction or magnitude of financial disturbances is more relevant for USD/VND volatility in a managed exchange-rate environment.

Two supplementary analyses are used to support the forecasting results without altering the validation-locked model. Dynamic HAC regressions examine whether lagged financial shocks are associated with the model-implied volatility path over the development sample. Weekly VAR/SVAR models provide additional evidence on short-run predictive relationships and dynamic shock transmission. These analyses are interpreted as supporting evidence rather than as the basis for model selection or structural causal claims.

The remainder of the paper is organized as follows. Section II reviews the literature on exchange-rate volatility, GARCH-family forecasting models, global and domestic financial shocks, and the Vietnamese exchange-rate market. Section III describes the data, forecasting-sample construction, model specifications, and evaluation framework. Section IV reports the preliminary diagnostics, validation-based model selection, final test-period results, and supplementary analyses. Section V discusses the main findings and concludes.

II. LITERATURE REVIEW

A. Volatility Modelling and Forecast Evaluation

The econometric foundation of conditional-volatility modelling begins with the ARCH model introduced by Engle [2]. Bollerslev [3] extended this framework into the GARCH model

by allowing conditional variance to depend on both past squared innovations and past conditional variance. These specifications capture volatility clustering, a common feature of financial returns in which periods of large movements tend to be followed by further large movements.

Subsequent models allow volatility to respond asymmetrically to past innovations. Nelson [4] proposed the EGARCH model, which expresses conditional variance in logarithmic form and allows positive and negative innovations of equal magnitude to have different effects. Glosten et al. [5] developed the GJR-GARCH specification, which captures asymmetry through an indicator term in the conditional-variance equation. These models provide relevant alternatives when the volatility response cannot be represented adequately by a symmetric GARCH process.

Model comparison should not rely exclusively on in-sample likelihood or information criteria. A more flexible volatility equation can fit historical data well without producing more accurate forecasts. Hansen and Lunde [6] compare a broad set of volatility models and find no evidence that more sophisticated specifications systematically outperform the GARCH(1,1) benchmark for exchange-rate returns. Their results establish the importance of benchmark discipline and formal out-of-sample evaluation.

A further difficulty is that latent conditional variance is not directly observable. Forecast accuracy must therefore be evaluated using a volatility proxy, such as squared returns or squared innovations. Patton [7] shows that noisy volatility proxies can distort model comparisons under unsuitable loss functions and identifies proxy-robust loss functions for volatility evaluation. This motivates the use of QLIKE as the primary forecast-ranking criterion in the present study. RMSE and MAE are reported as supplementary metrics.

Numerical differences in forecast loss do not necessarily imply statistically meaningful differences in predictive accuracy. Diebold and Mariano [8] develop tests of the null hypothesis that two competing forecasts have equal expected loss. This framework provides a basis for interpreting pairwise differences between the validation-locked model, GARCH-family alternatives, EGARCH- X specifications, and simple variance benchmarks.

These studies motivate a chronological forecasting design. The present study uses a training sample to select the conditional-mean equation and estimate candidate volatility models, a validation sample to compare recursive one-step-ahead forecasts and lock the preferred specification, and an untouched test sample for final out-of-sample evaluation.

B. Global and Domestic Financial Shocks

“latex Ozkaya and Altun [9] examine domestic and global causes of Turkish-lira volatility using GARCH models and Lyapunov exponents. Their study separates global financial conditions from domestic risk indicators and finds that exchange-rate volatility is related to variables including the VIX and credit-default-swap spreads. Their approach is relevant because

it treats exchange-rate volatility as the outcome of both internal dynamics and observable financial disturbances.

External financial conditions are also important in emerging Asian markets. Caporale et al. [10] investigate the effects of equity and bond portfolio inflows on exchange-rate volatility using monthly bilateral data for seven Asian emerging and developing economies. Their analysis combines GARCH models with Markov-switching specifications. The results support the broader view that emerging-market exchange rates can be sensitive to international capital allocation and changing global financial conditions.

C. Vietnam-Specific Evidence and Research Gap

Existing studies on Vietnam have focused mainly on the consequences of exchange-rate volatility for trade rather than on daily volatility forecasting. Thuy and Thuy [11] examine the impact of exchange-rate volatility on Vietnamese exports using quarterly data and an ARDL bounds-testing approach. Their results indicate that exchange-rate volatility negatively affects export volume in the long run.

Huynh and Hoang [12] study the effect of exchange-rate volatility on Vietnam's bilateral import value. Their analysis uses panel data and two volatility measures, including a GARCH-based measure and a moving standard deviation. This work confirms that exchange-rate volatility is relevant to Vietnam's external sector.

These studies establish the economic importance of exchange-rate volatility for Vietnam, but they do not address three forecasting questions. First, they do not compare GARCH-family models using a chronological train-validation-test framework. Second, they do not evaluate recursive one-step-ahead daily USD/VND volatility forecasts. Third, they do not test whether lagged global and domestic financial shocks improve forecast accuracy beyond the exchange rate's own volatility history.

The present study addresses this gap by comparing GARCH(1, 1), GJR-GARCH(1, 1), EGARCH(1, 1), custom no-(X) EGARCH, signed-shock EGARCH-X, and absolute-shock EGARCH-X specifications for daily USD/VND volatility forecasting. The forecasting analysis is complemented by development-sample HAC regressions and weekly recursive VAR analysis. These supplementary models are used to examine dynamic associations and predictive content; they do not alter the validation-locked forecasting model.

The contribution is therefore twofold. First, the study provides an out-of-sample comparison of standard and shock-augmented conditional-volatility models for USD/VND returns. Second, it evaluates whether lagged global and domestic financial shocks provide stable incremental forecasting information across model specifications, evaluation periods, and volatility proxies.

III. DATA AND METHODOLOGY

A. Data

The empirical analysis uses daily observations for the USD/VND nominal exchange rate and a set of global and

domestic financial variables. The raw sample covers the period from 4 January 2010 to 31 December 2025 and contains 4,021 observations. The dependent series is the USD/VND exchange rate, expressed as the number of Vietnamese dong per U.S. dollar. A higher exchange-rate value therefore indicates a depreciation of the Vietnamese dong.

The explanatory variables are selected to represent the main channels through which external and domestic financial conditions may affect USD/VND volatility. The CBOE Volatility Index (VIX) captures global risk aversion and financial-market uncertainty. The U.S. 10-year Treasury yield represents changes in global interest-rate conditions and the relative attractiveness of U.S. dollar-denominated assets. The U.S. Dollar Index (DXY) measures broad movements in the strength of the U.S. dollar against major currencies. WTI crude oil prices capture commodity-market shocks that may affect inflation expectations, trade conditions, and global risk sentiment. The VN-Index is included as a domestic financial-market variable reflecting local equity-market conditions.

These variables represent distinct but potentially interrelated financial channels. The purpose of the analysis is not to assume that global and domestic shocks are independent. Instead, the empirical question is whether lagged financial shocks contain incremental predictive information for USD/VND volatility after accounting for the persistence of the exchange-rate variance process.

The sample period includes several episodes of elevated market stress and policy adjustment, including the 2010 and 2011 exchange-rate adjustment periods, the COVID-19 shock, and the 2022–2023 U.S. monetary-tightening cycle. Including both tranquil and turbulent periods is necessary because volatility-forecasting models should be evaluated across different market regimes rather than only during relatively stable conditions.

The daily USD/VND log return is defined as

$$r_t^{FX} = 100 \ln \left(\frac{S_t}{S_{t-1}} \right), \quad (1)$$

where S_t denotes the USD/VND nominal exchange rate on day t . Returns are multiplied by 100 so that they are measured in percentage points.

B. Forecasting Sample Construction

Figure below summarizes the empirical workflow of the study. The analysis begins with the construction of a common daily dataset from raw market prices, including data cleaning, temporal alignment, return transformations, and the generation of global and domestic financial-shock variables. Preliminary diagnostics are then used to examine stationarity, distributional properties, serial dependence, and conditional heteroskedasticity. The model-ready sample is divided chronologically into training, validation, and untouched test periods. The training sample is used to select the ARMA conditional-mean specification and estimate the candidate volatility models, including GARCH, GJR-GARCH, EGARCH, custom no-(X) EGARCH, signed-shock EGARCH-X, and absolute-shock EGARCH-X specifications. Recursive one-step-ahead forecasts

are evaluated on the validation sample using QLIKE as the primary ranking criterion, with RMSE and MAE reported as supplementary metrics. The best-performing specification is then locked, re-estimated on the combined training and validation sample, and evaluated on the untouched test period. Robustness checks and supplementary EGARCH-X, HAC, and weekly recursive VAR analyses are conducted without altering the validation-based model-selection decision.

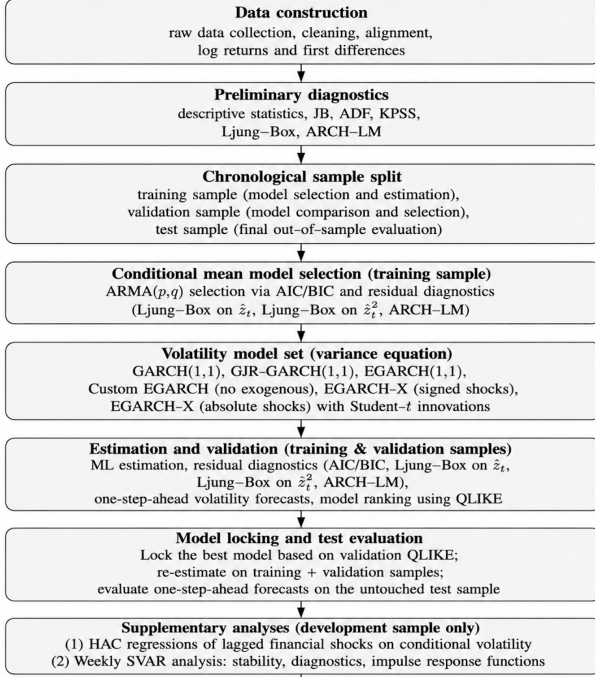


Fig. 1. Empirical workflow of the USD/VND volatility-forecasting framework.

The data-processing procedure is designed specifically for out-of-sample volatility forecasting. The raw data are first converted into numeric format, sorted chronologically, and checked for duplicated dates. The raw dataset contains no missing values and no duplicated observations. The sample is then divided chronologically into training, validation, and test periods before feature engineering is performed.

TABLE I
CHRONOLOGICAL SPLIT OF THE FORECASTING SAMPLE

Subsample	Period	Obs.	Purpose
Train	6 January 2010–31 December 2021	3,014	Model estimation
Validation	3 January 2022–29 December 2023	501	Model selection
Test	2 January 2024–31 December 2025	504	Final evaluation

The effective forecasting sample contains 4,019 observations. Two initial observations are unavailable because return construction and one-period lagged predictors require prior-day values.

The chronological split is imposed before constructing the forecasting variables. This prevents information from the validation or test periods from entering the training dataset.

At the boundaries between subsamples, a short trailing history from the preceding period is used only to calculate differences and lagged variables. No future observation is used to construct predictors for an earlier date.

The global and domestic financial shocks are transformed as follows:

$$\begin{aligned}
 \Delta VIX_t &= VIX_t - VIX_{t-1}, \\
 \Delta US10Y_t &= US10Y_t - US10Y_{t-1}, \\
 r_t^{DXY} &= 100 \ln \left(\frac{DXY_t}{DXY_{t-1}} \right), \\
 \Delta OIL_t &= OIL_t - OIL_{t-1}, \\
 r_t^{VNI} &= 100 \ln \left(\frac{VNI_t}{VNI_{t-1}} \right).
 \end{aligned} \tag{2}$$

WTI crude oil prices require a different transformation because the raw series contains a negative settlement price during the 2020 oil market disruption. A conventional log return is therefore not defined over the full sample. The oil shock is measured using the first difference in the WTI level:

$$\Delta OIL_t = OIL_t - OIL_{t-1}. \tag{3}$$

All financial shocks are lagged by one observation before entering the forecasting models:

$$X_{t-1} = \begin{bmatrix} \Delta VIX_{t-1} \\ \Delta US10Y_{t-1} \\ r_{t-1}^{DXY} \\ \Delta OIL_{t-1} \\ r_{t-1}^{VNI} \end{bmatrix}. \tag{4}$$

Lagging the shock variables ensures that the predictors are available before volatility for date t is forecast. This restriction is essential for a valid out-of-sample evaluation and is also economically plausible in a managed exchange-rate environment, where external shocks may be absorbed gradually rather than instantaneously.

The squared USD/VND return is retained as a robustness proxy for realized variance:

$$RV_t^{(r)} = (r_t^{FX})^2. \tag{5}$$

The realized-variance proxy is used only to evaluate volatility forecasts. It is not included as a predictor because its value becomes observable only after the return for date t has been realized.

TABLE II
VARIABLES USED IN THE VOLATILITY-FORECASTING DATASET

Variable	Construction	Role
fx_return	r_t^{FX}	USD/VND return series.
realized_var_proxy	$(r_t^{FX})^2$	Forecast-evaluation proxy.
vix_change_lag1	ΔVIX_{t-1}	Global risk-aversion shock.
us10y_change_lag1	$\Delta US10Y_{t-1}$	U.S. interest-rate shock.
dxy_return_lag1	r_{t-1}^{DXY}	U.S. dollar-strength shock.
oil_change_lag1	ΔOIL_{t-1}	Oil-market shock.
vnindex_return_lag1	r_{t-1}^{VNI}	Domestic equity-market shock.

Each GARCH-family specification is estimated under Normal, Student- t , and Skew- t innovations. For EGARCH, both symmetric and asymmetric variants are included in the candidate set.

The final forecasting datasets contain no duplicated dates, no overlapping observations across subsamples, and no missing values in the required modelling columns. The exported training, validation, and test datasets are used sequentially: candidate specifications are estimated on the training sample, model selection is based on validation-period forecasts, and the locked specification is evaluated once on the untouched test sample.

C. Conditional-Mean Model

Exchange-rate volatility is modelled after filtering predictable conditional-mean dynamics. The return series follows an ARMA(p, q) process:

$$r_t = \mu + \sum_{i=1}^p \phi_i r_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t, \quad (6)$$

where ε_t is the innovation after conditional-mean filtering.

Candidate ARMA specifications are estimated on the training sample only. Admissible models must converge under stationarity and invertibility constraints. Residual serial correlation is assessed using adjusted Ljung–Box tests. Among the admissible candidates, the model with the lowest Bayesian information criterion is selected:

$$BIC = -2\ell + k \ln(n), \quad (7)$$

where ℓ is the maximized log-likelihood, k is the number of estimated parameters, and n is the number of observations.

The selected ARMA order is locked before the volatility models are compared. The resulting innovations are used as the input series for conditional-variance estimation.

D. GARCH-Family Volatility Models

Let

$$\varepsilon_t = \sigma_t z_t, \quad h_t = \sigma_t^2, \quad (8)$$

where h_t is the conditional variance and z_t is a standardized innovation.

1) *GARCH Model*: The standard GARCH model extends the ARCH framework of [2] by allowing volatility to depend on both previous innovations and previous conditional variance [3]:

$$h_t = \omega + \alpha \varepsilon_{t-1}^2 + \beta h_{t-1}. \quad (9)$$

The ARCH coefficient α measures the immediate response to new information. The GARCH coefficient β measures volatility persistence.

2) *GJR-GARCH Model*: The GJR-GARCH model allows negative and positive innovations of the same magnitude to affect volatility differently [5]:

$$h_t = \omega + \alpha \varepsilon_{t-1}^2 + \gamma I(\varepsilon_{t-1} < 0) \varepsilon_{t-1}^2 + \beta h_{t-1}, \quad (10)$$

where $I(\varepsilon_{t-1} < 0)$ equals one when the previous innovation is negative and zero otherwise. The coefficient γ captures the additional effect of a negative innovation.

3) *EGARCH Model*: The EGARCH model expresses conditional variance in logarithmic form [4]:

$$\ln(h_t) = \omega + \alpha (|z_{t-1}| - c_0) + \gamma z_{t-1} + \beta \ln(h_{t-1}), \quad (11)$$

where

$$c_0 = \sqrt{\frac{2}{\pi}}. \quad (12)$$

The logarithmic specification ensures positive conditional variance without requiring all coefficients to be non-negative. The magnitude term captures the effect of large innovations, while the signed term captures asymmetric responses.

The centering constant c_0 is fixed explicitly in the implementation. It is not estimated from the data.

E. EGARCH-X Models with Financial Shocks

The EGARCH-X model extends the EGARCH variance recursion by incorporating lagged standardized financial shocks. This extension tests whether global and domestic variables contain predictive information beyond the exchange rate's own volatility history.

The general custom EGARCH-X specification is

$$\ln(h_t) = \omega + \alpha (|z_{t-1}| - c_0) + \gamma z_{t-1} + \beta \ln(h_{t-1}) + \sum_{j=1}^J \delta_j q(Z_{j,t-1}). \quad (13)$$

Five custom EGARCH-X specifications are estimated on the same aligned sample. The first specification is a custom EGARCH benchmark without external regressors. The second adds the broad crisis-period control. The third includes both the crisis-period dummy and the fixed one-day devaluation dummy. The final two specifications extend the event-control model by adding either lagged signed financial shocks or lagged absolute financial shocks.

Let

$$\mathbf{D}_t = \begin{bmatrix} D_t^{crisis} \\ D_t^{dev} \end{bmatrix}, \quad (14)$$

where D_t^{crisis} captures the predefined crisis periods and D_t^{dev} is the fixed one-day dummy for the 11 February 2010 devaluation event. The shock transformation is defined as

$$q(Z_{j,t-1}) = \begin{cases} 0, & \text{benchmark and event-control models,} \\ Z_{j,t-1}, & \text{signed-shock EGARCH-X,} \\ |Z_{j,t-1}|, & \text{absolute-shock EGARCH-X.} \end{cases} \quad (15)$$

Three directly comparable custom EGARCH specifications are estimated:

- 1) custom EGARCH without external regressors;
- 2) signed-shock EGARCH-X with lagged standardized financial shocks;
- 3) absolute-shock EGARCH-X with lagged absolute standardized financial shocks.

The custom EGARCH model without external regressors serves as the benchmark for evaluating the incremental predictive content of financial shocks. The signed-shock specification tests whether the direction of lagged global and domestic financial shocks contains information for forecasting subsequent USD/VND volatility. The absolute-shock specification tests whether the magnitude of these shocks matters regardless of direction.

All three custom specifications are estimated using the same likelihood function, aligned estimation sample, initialization procedure, parameter constraints, and recursive one-step-ahead forecasting protocol. This common-sample design ensures that differences in model fit and forecast performance reflect the additional information contained in the lagged financial-shock variables rather than differences in sample composition or estimation settings.

F. Maximum-Likelihood Estimation

The parametric volatility models are estimated under standardized Student- t innovations to accommodate heavy-tailed exchange-rate movements:

$$z_t \sim t_\nu^*, \quad \nu > 2. \quad (16)$$

For the custom EGARCH-X specifications, the observation-level log-likelihood is

$$\begin{aligned} \ell_t = & \ln \Gamma\left(\frac{\nu+1}{2}\right) - \ln \Gamma\left(\frac{\nu}{2}\right) \\ & - \frac{1}{2} \ln[\pi(\nu-2)] - \frac{1}{2} \ln(h_t) \\ & - \frac{\nu+1}{2} \ln\left[1 + \frac{\varepsilon_t^2}{h_t(\nu-2)}\right]. \end{aligned} \quad (17)$$

The parameter vector is estimated by maximizing

$$\mathcal{L}(\theta) = \sum_{t=1}^T \ell_t. \quad (18)$$

The custom EGARCH-X implementation imposes the stability restriction

$$|\beta| < 1. \quad (19)$$

Because the model is specified in log variance, positivity follows from

$$h_t = \exp[\ln(h_t)] > 0. \quad (20)$$

The recursion is initialized using an exponentially weighted backcast based on early squared innovations. Multiple admissible starting values are used during bounded numerical optimization. A fitted model is rejected when it produces a non-finite likelihood, non-finite parameters, or non-positive conditional variance.

G. Benchmark Forecasts

The parametric models are compared with three non-parametric benchmarks: exponentially weighted moving-average variance, expanding historical variance, and rolling-window variance.

Their one-step-ahead forecasts are

$$\begin{aligned} \hat{h}_{t|t-1}^{EWMA} &= \lambda \hat{h}_{t-1|t-2}^{EWMA} + (1-\lambda) \varepsilon_{t-1}^2, \\ \hat{h}_{t|t-1}^{Hist} &= \frac{1}{t-1} \sum_{s=1}^{t-1} \varepsilon_s^2, \\ \hat{h}_{t|t-1}^{Roll} &= \frac{1}{L} \sum_{s=t-L}^{t-1} \varepsilon_s^2, \end{aligned} \quad (21)$$

where λ is the EWMA decay parameter and L is the rolling-window length. Both values are fixed before the untouched test period is evaluated.

H. Recursive One-Step-Ahead Forecasting

Validation and test forecasts are generated recursively at a one-day horizon. For model m , the generic forecast recursion is

$$\hat{h}_{t|t-1}^{(m)} = g_m(\varepsilon_{t-1}, \hat{h}_{t-1|t-2}^{(m)}, \mathbf{x}_{t-1}; \hat{\theta}_m). \quad (22)$$

The estimated parameters remain fixed throughout each evaluation window. After the forecast for date t is generated, the newly observed innovation ε_t is incorporated into the recursion used to forecast date $t+1$. Future returns and future shocks are never used to generate a current forecast.

I. Volatility Proxies and Forecast Evaluation

Conditional variance is latent. The primary volatility proxy is the squared innovation from the locked ARMA mean equation:

$$RV_t^{(\varepsilon)} = \varepsilon_t^2. \quad (23)$$

A robustness evaluation uses the squared raw exchange-rate return:

$$RV_t^{(r)} = r_t^2. \quad (24)$$

The primary evaluation criterion is QLIKE loss:

$$QLIKE_t^{(m)} = \ln\left(\hat{h}_{t|t-1}^{(m)}\right) + \frac{RV_t}{\hat{h}_{t|t-1}^{(m)}}. \quad (25)$$

Lower QLIKE values indicate more accurate volatility forecasts. QLIKE is used as the primary ranking criterion because it is robust to measurement error in the volatility proxy under standard conditions [7].

Additional forecast-error measures are

$$\begin{aligned} MSE_m &= \frac{1}{T} \sum_{t=1}^T \left(RV_t - \hat{h}_{t|t-1}^{(m)} \right)^2, \\ RMSE_m &= \sqrt{MSE_m}, \\ MAE_m &= \frac{1}{T} \sum_{t=1}^T \left| RV_t - \hat{h}_{t|t-1}^{(m)} \right|. \end{aligned} \quad (26)$$

The final model is selected exclusively by validation QLIKE:

$$m^* = \arg \min_{m \in \mathcal{M}} \overline{QLIKE}_m^{Valid}, \quad (27)$$

where $\mathcal{M}$ contains the GARCH-family models, custom EGARCH-X models, and benchmark forecasts. MSE is used only as a tie-breaking criterion when validation QLIKE values are effectively identical.

After the final model m^* has been locked, the required models are re-estimated using the combined training and validation period. Their recursive forecasts are then evaluated on the untouched test period. The test results are reported but are not used to change the selected model.

J. Residual Diagnostics

For each fitted volatility model, standardized residuals are calculated as

$$\hat{z}_t = \frac{\hat{\varepsilon}_t}{\sqrt{\hat{h}_t}}. \quad (28)$$

The Ljung–Box test is applied to both $\hat{z}_t$ and $\hat{z}_t^2$. The ARCH-LM test is applied to the standardized residuals. These diagnostics identify remaining serial dependence or conditional heteroskedasticity. They are reported as model-quality checks but do not override the validation-based selection rule.

K. Supplementary Dynamic HAC Regressions

The first supplementary analysis evaluates whether lagged financial shocks are associated with the estimated volatility path of the validation-locked model. The untouched test period is excluded.

Let

$$v_t = \ln(\hat{\sigma}_t^*), \quad (29)$$

where $\hat{\sigma}_t^*$ is the conditional standard deviation generated by the locked model after re-estimation on the development sample.

The baseline dynamic regression is

$$v_t = c + \rho v_{t-1} + \kappa D_{t-1} + u_t, \quad (30)$$

where D_{t-1} is an optional lagged crisis indicator defined before model estimation.

The signed-shock specification is

$$v_t = c + \rho v_{t-1} + \sum_{j=1}^J \beta_j Z_{j,t-1} + \kappa D_{t-1} + u_t. \quad (31)$$

The absolute-shock specification is

$$v_t = c + \rho v_{t-1} + \sum_{j=1}^J \lambda_j |Z_{j,t-1}| + \kappa D_{t-1} + u_t. \quad (32)$$

The regressions are estimated by ordinary least squares with HAC standard errors. Their coefficients describe development-sample associations. They are not interpreted as structural causal effects and are not used for forecasting model selection.

L. Supplementary Weekly SVAR

The second supplementary analysis examines short-run transmission using a weekly recursively identified structural VAR. Contemporaneous daily shocks are aggregated to weekly frequency. Weekly shocks and exchange-rate returns are summed within each week. Log conditional volatility is averaged within each week.

The weekly variable vector is

$$\mathbf{Y}_\tau = \begin{bmatrix} \Delta VIX_\tau^{(w)} \\ \Delta US10Y_\tau^{(w)} \\ r_\tau^{DXY,(w)} \\ \Delta OIL_\tau^{(w)} \\ r_\tau^{VNI,(w)} \\ Target_\tau^{(w)} \end{bmatrix}. \quad (33)$$

The target is estimated separately as weekly USD/VND return and weekly log conditional volatility.

The reduced-form VAR is

$$\mathbf{Y}_\tau = \mathbf{c} + \sum_{\ell=1}^p \mathbf{A}_\ell \mathbf{Y}_{\tau-\ell} + \mathbf{u}_\tau. \quad (34)$$

The lag order is selected by minimizing BIC among stable candidates:

$$p^* = \arg \min_{p \in \mathcal{P}_{stable}} BIC(p). \quad (35)$$

If no stable level specification is available for weekly log volatility, the permitted fallback target is

$$\Delta v_\tau^{(w)} = v_\tau^{(w)} - v_{\tau-1}^{(w)}. \quad (36)$$

Structural shocks are identified using the recursive ordering

$$\begin{aligned} VIX &\rightarrow US10Y \rightarrow DXY \\ &\rightarrow OIL \rightarrow VNINDEX \rightarrow Target. \end{aligned} \quad (37)$$

The ordering assumes that global financial conditions affect the more locally specific variables contemporaneously, while

the USD/VND target responds last within the week. Orthogonalized impulse-response functions and Granger-causality tests are reported as supplementary evidence. The weekly SVAR does not alter the validation-locked forecasting model.

IV. RESULTS

A. Preliminary Time-Series Diagnostic Results

This subsection reports preliminary full-sample diagnostics for the processed daily series. These results are used to characterize the statistical properties of the data and to motivate the subsequent model specifications. They are not used to select the final forecasting model. The conditional-mean order and the volatility specification are selected separately within the time-ordered training and validation framework, while the test sample remains untouched until the final out-of-sample evaluation.

1) *Stationarity and distributional properties:* Table III reports the Augmented Dickey–Fuller (ADF), Kwiatkowski–Phillips–Schmidt–Shin (KPSS), and Jarque–Bera (JB) test results. The ADF test evaluates the null hypothesis of a unit root, whereas the KPSS test evaluates the null hypothesis of stationarity. For every transformed variable, the ADF null hypothesis is rejected at the 1% significance level. Conversely, the KPSS null hypothesis cannot be rejected. The transformed daily series are therefore treated as stationary and no additional differencing is required before estimation.

TABLE III
STATIONARITY AND NORMALITY DIAGNOSTICS

Variable	ADF statistic	ADF p -value	KPSS statistic	KPSS p -value	JB statistic	JB p -value
r_t^{FX}	-16.250	< 0.001	0.126	> 0.100	7,460,584.931	< 0.001
ΔVIX_t	-15.237	< 0.001	0.009	> 0.100	144,656.692	< 0.001
$\Delta US10Y_t$	-47.290	< 0.001	0.283	> 0.100	434.770	< 0.001
r_t^{DXY}	-62.551	< 0.001	0.083	> 0.100	391.297	< 0.001
ΔOIL_t	-10.903	< 0.001	0.046	> 0.100	10,365,767.446	< 0.001
r_t^{VNI}	-41.677	< 0.001	0.095	> 0.100	3,395.068	< 0.001

Notes: The ADF test has the null hypothesis of a unit root. The KPSS test has the null hypothesis of stationarity. KPSS p -values are reported as > 0.100 because the test statistics lie below the lower boundary of the available lookup table. The JB test has the null hypothesis of normality. The ADF effective sample size varies slightly across variables because of variable-specific missing-value trimming and automatic lag selection.

Although the transformed variables are stationary, the JB test strongly rejects normality for every series. The non-normality is particularly pronounced for the USD/VND return series. The daily USD/VND return has a mean of 0.0088, a standard deviation of 0.2125, a minimum of -2.3942, and a maximum of 6.2139. Its skewness is 6.4754, while its excess kurtosis reaches 210.9132. These results indicate a highly right-skewed and heavy-tailed return distribution. This feature supports the use of volatility models that allow for non-Gaussian innovations rather than relying on a normal-error specification.

2) *Serial correlation and volatility clustering:* The Ljung–Box results in Table IV reveal statistically significant serial dependence in the USD/VND return series. The null hypothesis of no autocorrelation is rejected at lags 5, 10, and 20. The sample autocorrelation and partial autocorrelation functions also contain several statistically significant short-run coefficients. In particular, the return autocorrelation function is significant at

lags 1, 3–5, 8–10, 12–13, and 18. The partial autocorrelation function is significant at lags 1–5, 8, 10, and 12. These patterns motivate the inclusion of an ARMA conditional-mean component. However, the final ARMA order is determined through the formal training-sample model-selection procedure rather than visual inspection of the correlograms alone.

TABLE IV
SERIAL DEPENDENCE AND VOLATILITY-CLUSTERING DIAGNOSTICS

Series	$Q(5)$	p -value	$Q(10)$	p -value	$Q(20)$	p -value
r_t^{FX}	109.191	< 0.001	147.410	< 0.001	181.743	< 0.001
$ r_t^{FX} $	1,015.330	< 0.001	1,495.365	< 0.001	1,694.245	< 0.001
$(r_t^{FX})^2$	49.092	< 0.001	83.854	< 0.001	90.620	< 0.001

Notes: $Q(k)$ denotes the Ljung–Box statistic evaluated at lag k . The null hypothesis is that the series contains no autocorrelation up to the reported lag. The tests are applied separately to daily USD/VND returns, absolute returns, and squared returns.

The evidence of conditional heteroskedasticity is reinforced by the ARCH–LM test. Using 10 lags, the ARCH–LM statistic is 59.512 with a p -value below 0.001. The null hypothesis of no ARCH effects is therefore rejected. Moreover, the Ljung–Box statistics for both absolute and squared USD/VND returns are statistically significant at all reported lags. The persistence is especially strong for absolute returns, for which the $Q(10)$ statistic reaches 1,495.365. These results confirm substantial volatility clustering and justify the use of GARCH-family models for conditional-volatility forecasting.

B. Chronological Sample Split and Data Integrity

The model-ready daily sample contains 4,019 observations after the lagged shock variables required by the forecasting pipeline are aligned. The observations are divided chronologically into a training sample, a validation sample, and an untouched test sample. The training period is used to select the conditional-mean specification and estimate the candidate volatility models. The validation period is used exclusively to compare one-step-ahead volatility forecasts and lock the final forecasting specification. The test period remains excluded from all model-selection decisions and is used only for the final out-of-sample reporting exercise.

TABLE V
CHRONOLOGICAL SAMPLE SPLIT AND DATA-INTEGRITY AUDIT

Sample	Period	Obs.	Missing values	Non-finite values	Duplicate dates	Zero-return share
Training	2010-01-06–2021-12-31	3,014	0	0	0	0.1543
Validation	2022-01-03–2023-12-29	501	0	0	0	0.1178
Test	2024-01-02–2025-12-31	504	0	0	0	0.1190

Note: All samples are sorted chronologically and contain no overlapping dates. The required modelling inputs contain no missing or non-finite values. The test sample is excluded from conditional-mean selection, volatility-model selection, hyperparameter comparison, EGARCH-X interpretation, HAC regressions, and weekly VAR/SVAR analysis.

The chronology audit confirms that the training sample ends on 2021-12-31, the validation sample begins on 2022-01-03 and ends on 2023-12-29, and the test sample begins on 2024-01-02. The strict temporal separation prevents future observations from entering the model-selection stage.

C. ARMA Conditional-Mean Selection on the Training Sample

The conditional-mean equation is selected exclusively on the training sample. A grid of ARMA(p, q) specifications with $p, q \in \{0, \dots, 5\}$ is estimated under stationarity and invertibility constraints. A converged candidate is retained as diagnostically adequate only if its adjusted Ljung–Box residual p -values are at least 0.05 at lags 10, 20, and 30. The model with the lowest BIC is then selected among the specifications that satisfy this residual-whiteness requirement.

TABLE VI
TRAINING-SAMPLE ARMA CONDITIONAL-MEAN SELECTION

Model	AIC	BIC	LB(10)	LB(20)	LB(30)	ARCH-LM (10)	Diag. pass	Sel.
ARMA(4,5)	-676.521	-610.400	0.0719	0.0584	0.1978	0.0017	Yes	Yes
ARMA(2,3)	-664.005	-621.928	0.1559	0.0263	0.1591	0.0018	No	No
ARMA(3,2)	-662.992	-620.915	0.0875	0.0217	0.1477	0.0022	No	No
ARMA(3,3)	-662.324	-614.236	0.0790	0.0172	0.1232	0.0018	No	No
ARMA(5,4)	-677.050	-610.929	0.0274	0.0619	0.2310	0.0024	No	No

Note: LB(k) denotes the adjusted Ljung–Box residual p -value at lag k . ARCH-LM(10) denotes the ARCH-LM p -value at lag 10. Diag. pass requires convergence and adjusted Ljung–Box residual p -values above 0.05 at lags 10, 20, and 30. The ARCH-LM result is reported as evidence of remaining conditional heteroskedasticity and does not determine conditional-mean admissibility. The complete ARMA grid, including non-converged candidates, is reported in the Appendix.

ARMA(4,5) is the only estimated specification that satisfies the adjusted residual-whiteness requirement at all three reported lags. Although several lower-order alternatives have lower BIC values, they retain statistically significant residual serial dependence and are therefore rejected. The ARCH–LM p -value for the selected ARMA(4,5) residuals remains below 0.01. The conditional-mean equation absorbs the relevant linear dependence, but conditional heteroskedasticity remains. The ARMA(4,5) innovations are therefore used as the inputs for volatility modeling.

D. Volatility-Model Selection on the Validation Sample

After the ARMA(4,5) conditional mean is fixed, six Student- t conditional-volatility specifications are estimated on the training sample: GARCH(1,1), GJR-GARCH(1,1), EGARCH(1,1), a custom EGARCH model without external regressors, a signed-shock EGARCH-X model, and an absolute-shock EGARCH-X model. The EGARCH-X specifications use standardized lagged shock variables. Three non-parametric benchmarks are also evaluated: EWMA(0.94), expanding historical variance, and 22-day rolling variance.

All six estimated volatility specifications converge on the training sample. The standardized-residual diagnostics are satisfactory for all six Student- t candidates. The custom EGARCH-X models also pass the parameter audit: the absolute persistence coefficients satisfy $|\beta| < 0.999$, the Student- t degrees-of-freedom parameters exceed 2.05, and all estimated parameters are finite.

Forecast accuracy is evaluated using recursive one-step-ahead forecasts. The primary observable variance proxy is the squared recursive ARMA innovation,

$$RV_t^{\text{main}} = \hat{\varepsilon}_t^2, \quad (38)$$

where $\hat{\varepsilon}_t$ is the one-step-ahead innovation from the selected ARMA(4,5) conditional-mean model. The primary ranking criterion is the QLIKE loss:

$$\text{QLIKE}_t = \ln \left(\hat{h}_{t|t-1} \right) + \frac{RV_t^{\text{main}}}{\hat{h}_{t|t-1}}. \quad (39)$$

where $\hat{h}_{t|t-1}$ denotes the forecast conditional variance. Lower QLIKE values indicate better forecast performance. RMSE and MAE are reported as supplementary measures.

TABLE VII
VALIDATION-SAMPLE VOLATILITY-FORECAST ACCURACY

Rank	Model	QLIKE	RMSE	MAE	Mean forecast variance	Locked
1	GJR-GARCH(1,1)-t	-2.6783	0.1747	0.0536	0.0433	Yes
2	EGARCH-X absolute lag-1- t	-2.6684	0.1722	0.0610	0.0552	No
3	EGARCH-X signed lag-1- t	-2.6046	0.1695	0.0535	0.0434	No
4	Custom EGARCH no- X - t	-2.5695	0.1698	0.0535	0.0432	No
5	EGARCH(1,1)- t	-2.5694	0.1698	0.0534	0.0432	No
6	EWMA(0.94)	-2.4185	0.1730	0.0543	0.0423	No
7	GARCH(1,1)- t	-2.3967	0.1687	0.0453	0.0293	No
8	Historical variance	-2.1712	0.1765	0.0620	0.0460	No
9	Rolling variance (22 days)	-2.0779	0.1795	0.0567	0.0423	No

Note: The validation sample covers 3 January 2022–29 December 2023 and contains 501 recursive one-step-ahead forecasts. The primary volatility proxy is the squared recursive ARMA innovation defined in Equation (38). QLIKE is the model-selection criterion; lower values indicate better forecast accuracy. GJR-GARCH(1,1)- t is locked because it achieves the lowest validation QLIKE.

GJR-GARCH(1,1)- t achieves the lowest validation QLIKE of -2.6783 and is therefore locked as the final forecasting specification. The absolute-shock EGARCH-X model ranks second with a QLIKE of -2.6684 , followed by the signed-shock EGARCH-X model with a QLIKE of -2.6046 . The validation results show that the magnitude of global and domestic shocks can contain useful supplementary forecasting information. However, the improvement is not sufficient to replace the validation-selected GJR-GARCH specification.

The validation ranking differs from the in-sample likelihood ranking. The EGARCH-type specifications obtain stronger in-sample likelihood values, whereas GJR-GARCH(1,1)- t delivers the lowest validation QLIKE. The forecasting model is therefore selected using out-of-sample validation accuracy rather than in-sample fit.

E. Locked-Model Evaluation on the Untouched Test Sample

After the validation stage is completed, the ARMA(4,5) conditional-mean model and the candidate volatility models are re-estimated using the combined training and validation sample. This development sample contains 3,515 observations. The untouched 2024–2025 test sample contains 504 observations and is evaluated only after the locked-model decision has been finalized.

The refitted locked GJR-GARCH(1,1)- t specification remains diagnostically adequate. Table VIII reports the relevant standardized-residual diagnostics.

TABLE VIII
DEVELOPMENT-SAMPLE DIAGNOSTICS FOR THE LOCKED GJR-GARCH
MODEL

Model	LB(10)	LB(20)	LB ² (10)	LB ² (20)	ARCH-LM (10)	Diag. pass
GJR-GARCH(1,1)- <i>t</i>	0.2909	0.6156	0.0904	0.6572	0.0878	Yes

Note: LB(*k*) and LB²(*k*) denote Ljung–Box *p*-values for standardized residuals and squared standardized residuals, respectively. ARCH-LM(10) denotes the ARCH-LM *p*-value at lag 10. The diagnostics are computed after re-estimating the locked ARMA(4,5)–GJR-GARCH(1,1)-*t* model on the combined training and validation sample.

The standardized-residual Ljung–Box *p*-values are 0.2909 and 0.6156 at lags 10 and 20, while the corresponding squared-standardized-residual *p*-values are 0.0904 and 0.6572. The ARCH–LM *p*-value is 0.0878. Residual heteroskedasticity is therefore not detected at the 5% significance level after the development-sample refit.

TABLE IX
LOCKED-MODEL EVALUATION ON THE UNTOUCHED TEST SAMPLE

Rank	Model	QLIKE	RMSE	MAE	Mean forecast variance	Status
1	EGARCH(1,1)- <i>t</i>	-2.8511	0.0894	0.0425	0.0406	Comp.
2	Custom EGARCH no- <i>X</i> - <i>t</i>	-2.8510	0.0894	0.0426	0.0407	Comp.
3	EGARCH-X absolute lag-1- <i>t</i>	-2.8485	0.0936	0.0433	0.0410	Comp.
4	EGARCH-X signed lag-1- <i>t</i>	-2.8482	0.0892	0.0435	0.0424	Comp.
5	GARCH(1,1)- <i>t</i>	-2.8059	0.0859	0.0377	0.0326	Comp.
6	GJR-GARCH(1,1)-<i>t</i>	-2.7751	0.0887	0.0381	0.0325	Locked
7	EWMA(0.94)	-2.7594	0.0865	0.0375	0.0305	Bench.
8	Rolling variance (22 days)	-2.6962	0.0884	0.0381	0.0301	Bench.
9	Historical variance	-2.4523	0.0889	0.0499	0.0449	Bench.

Note: The test sample covers 2 January 2024–31 December 2025 and contains 504 recursive one-step-ahead forecasts. GJR-GARCH(1,1)-*t* was selected using validation QLIKE and locked before the untouched test sample was evaluated. The remaining specifications are reporting-only comparators. Test-sample ranks are descriptive and are not used for model reselection. Comp. denotes comparator; Bench. denotes benchmark.

The locked GJR-GARCH(1,1)-*t* model remains competitive on the untouched test sample. Its QLIKE of -2.7751 is numerically lower than the values obtained by EWMA(0.94), 22-day rolling variance, and expanding historical variance. However, it does not dominate the EGARCH-family comparators during the 2024–2025 test period. The standard EGARCH(1,1)-*t* specification obtains the lowest test QLIKE of -2.8511 , while the custom no-*X* EGARCH and both EGARCH-X models also rank above the locked specification.

This result does not justify ex-post model replacement. Replacing the locked model after inspecting the test ranking would use the test sample as a second validation period and invalidate the forecasting design. The result instead shows that the relative forecast performance of conditional-volatility models varies across evaluation periods. GJR-GARCH dynamics perform best during the 2022–2023 validation period, whereas EGARCH-type dynamics are more effective during the 2024–2025 test period.

F. Robustness Analysis

Two robustness exercises are conducted. The first replaces the primary innovation-based variance proxy with squared daily USD/VND returns. The second examines the sensitivity of the standard GARCH-family rankings to the assumed innovation

distribution. These exercises do not reopen the locked-model decision.

1) *Alternative Variance Proxy*: The alternative proxy is defined as

$$RV_t^{\text{robust}} = (r_t^{FX})^2. \quad (40)$$

TABLE X
UNTOUCHED-TEST ROBUSTNESS RESULTS USING SQUARED USD/VND
RETURNS

Rank	Model	QLIKE	RMSE	MAE	Locked
1	EGARCH-X absolute lag-1- <i>t</i>	-2.8973	0.0921	0.0427	No
2	EGARCH(1,1)- <i>t</i>	-2.8947	0.0879	0.0423	No
3	Custom EGARCH no- <i>X</i> - <i>t</i>	-2.8946	0.0879	0.0423	No
4	EGARCH-X signed lag-1- <i>t</i>	-2.8943	0.0875	0.0432	No
5	GARCH(1,1)- <i>t</i>	-2.8470	0.0842	0.0376	No
6	GJR-GARCH(1,1)-<i>t</i>	-2.8202	0.0868	0.0376	Yes
7	EWMA(0.94)	-2.8098	0.0848	0.0374	No
8	Rolling variance (22 days)	-2.7636	0.0868	0.0380	No
9	Historical variance	-2.4797	0.0873	0.0498	No

Note: The alternative volatility proxy is the squared daily USD/VND return defined in Equation (40). The untouched test sample contains 504 recursive one-step-ahead forecasts. Lower QLIKE, RMSE, and MAE values indicate better forecast accuracy. Test-sample ranks are descriptive only. The robustness exercise does not reopen the model-selection stage.

On the untouched test sample, the absolute-shock EGARCH-X specification ranks third under the primary innovation-based proxy and first under the squared-return robustness proxy. Its QLIKE is -2.8973 . The locked GJR-GARCH(1,1)-*t* model ranks sixth but continues to obtain a numerically lower QLIKE than the three simple variance benchmarks. The results indicate that the relative performance of shock-augmented EGARCH models is sensitive to the volatility proxy used for evaluation.

2) *Innovation-Distribution Sensitivity*: The main candidate set uses Student-*t* innovations as an ex-ante specification because the preliminary diagnostics indicate pronounced heavy tails in USD/VND returns. As a sensitivity exercise, the standard GARCH(1,1), GJR-GARCH(1,1), and EGARCH(1,1) specifications are re-estimated under Gaussian, Student-*t*, and skewed Student-*t* innovations. The distributional variants are evaluated on the validation sample after the core Student-*t* comparison has been defined.

TABLE XI
VALIDATION-SAMPLE SENSITIVITY ACROSS INNOVATION DISTRIBUTIONS

Rank	Model	QLIKE	RMSE	MAE
1	GJR-GARCH(1,1)-Normal	-2.6984	0.1776	0.0577
2	GJR-GARCH(1,1)-Student- <i>t</i>	-2.6783	0.1747	0.0536
2	GJR-GARCH(1,1)-Skew- <i>t</i>	-2.6783	0.1747	0.0536
4	GARCH(1,1)-Normal	-2.6077	0.1707	0.0532
5	GARCH(1,1)-Skew- <i>t</i>	-2.5979	0.1708	0.0536
6	EGARCH(1,1)-Skew- <i>t</i>	-2.5901	0.1711	0.0572
7	EGARCH(1,1)-Student- <i>t</i>	-2.5694	0.1698	0.0534
8	GARCH(1,1)-Student- <i>t</i>	-2.3967	0.1687	0.0453
9	EGARCH(1,1)-Normal	-2.3872	0.1798	0.0603

Note: The sensitivity analysis uses the 501-observation validation sample. Lower QLIKE, RMSE, and MAE values indicate better forecast accuracy. GJR-GARCH(1,1)-Student-*t* and GJR-GARCH(1,1)-Skew-*t* produce identical reported forecast metrics and therefore share the same rank. This robustness exercise does not reopen the locked-model decision, which is based on the predefined Student-*t* candidate set.

The Gaussian GJR-GARCH(1, 1) specification obtains the lowest QLIKE of -2.6984 . The Student- t and skewed Student- t GJR-GARCH specifications are next and return identical validation metrics to the reported precision. The results indicate that the preference for GJR-GARCH dynamics is not confined to a single innovation distribution. At the same time, the point ranking is sensitive to the distributional assumption. The Gaussian variant is therefore reported as a robustness result rather than used to revise the previously locked Student- t specification.

G. Supplementary Shock Analysis

The forecasting analysis above determines the locked volatility model. The following development-sample analyses examine how global and domestic financial shocks relate to USD/VND volatility and weekly exchange-rate dynamics. These results are supplementary. The untouched test sample remains excluded from EGARCH-X interpretation, HAC regressions, and weekly recursive VAR estimation.

1) *EGARCH-X Coefficient Patterns*: Table XII reports the development-sample shock coefficients from the signed-shock and absolute-shock EGARCH-X specifications. The regressors are standardized lagged shocks. The coefficients are interpreted descriptively because the exported output contains point estimates but does not report standard errors or p -values.

TABLE XII
DEVELOPMENT-SAMPLE EGARCH-X SHOCK-COEFFICIENT ESTIMATES

Lagged standardized shock	Signed-shock EGARCH-X	Absolute-shock EGARCH-X
$z_{t-1}^{\Delta VIX}$ or $ z_{t-1}^{\Delta VIX} $	0.0342	-0.0199
$z_{t-1}^{\Delta US10Y}$ or $ z_{t-1}^{\Delta US10Y} $	-0.0122	0.0140
z_{t-1}^{DXY} or $ z_{t-1}^{DXY} $	0.0455	0.0440
$z_{t-1}^{\Delta OIL}$ or $ z_{t-1}^{\Delta OIL} $	-0.0534	0.0180
z_{t-1}^{VNI} or $ z_{t-1}^{VNI} $	0.0066	0.0497
Persistence coefficient β	0.9763	0.9566

Note: The signed-shock specification includes lagged standardized financial shocks, while the absolute-shock specification includes their absolute magnitudes. Both EGARCH-X models are supplementary comparators rather than the validation-locked forecasting model. The table reports point estimates only; statistical significance is not assessed because standard errors are not available in the exported coefficient output.

Both EGARCH-X specifications imply persistent volatility dynamics. The estimated persistence coefficient is 0.9763 in the signed-shock model and 0.9566 in the absolute-shock model. Standardized DXY movements enter positively in both specifications. In the signed-shock model, the oil-price change has the largest coefficient in absolute value and enters negatively. In the absolute-shock model, the magnitude of VNINDEX returns has the largest positive coefficient, followed by the magnitude of DXY returns. These point estimates suggest that the informational role of financial shocks depends on whether shock direction or shock magnitude is modeled. They do not establish statistical significance or causal effects.

2) *Dynamic HAC Shock-Volatility Regressions*: Dynamic HAC regressions are estimated using the development-sample conditional volatility series generated by the locked

ARMA(4, 5)-GJR-GARCH(1, 1)- t model. The dependent variable is $\log(\hat{\sigma}_t)$. The baseline specification controls for lagged log conditional volatility. The signed-shock and absolute-shock specifications then add standardized lagged financial shocks. Newey-West standard errors are computed using five lags. The untouched test period is excluded from all HAC estimations.

TABLE XIII
DYNAMIC HAC REGRESSIONS FOR SELECTED CONDITIONAL VOLATILITY

Variable	Signed coef.	Signed p -value	Absolute coef.	Absolute p -value
Constant	-0.1966	< 0.001	-0.1996	< 0.001
$\log(\hat{\sigma}_{t-1})$	0.8974	< 0.001	0.8968	< 0.001
$z_{t-1}^{\Delta VIX}$	-0.0031	0.2401	-0.0051	0.1080
$z_{t-1}^{\Delta US10Y}$	-0.0016	0.5953	-0.0009	0.8327
z_{t-1}^{DXY}	-0.0052	0.0735	-0.0031	0.5165
$z_{t-1}^{\Delta OIL}$	-0.0005	0.8132	0.0006	0.8487
z_{t-1}^{VNI}	-0.0008	0.7990	0.0106	0.0116

Note: The dependent variable is $\log(\hat{\sigma}_t)$, where $\hat{\sigma}_t$ is generated by the development-sample ARMA(4,5)-GJR-GARCH(1,1)- t model. The signed specification includes lagged standardized shocks. The absolute specification includes the corresponding absolute shock magnitudes, $|z_{t-1}|$. HAC standard errors are computed using five lags. Bold values identify the statistically significant financial-shock estimate at the 5% level.

TABLE XIV
DYNAMIC HAC MODEL SUMMARY

Model	Obs.	R^2	ΔR^2	Adj. R^2	AIC	BIC	LB(10)	LB(20)
Lagged-volatility control	3,514	0.8079	–	0.8078	-2,804.24	-2,791.91	0.3449	0.0081
Signed shocks	3,514	0.8082	0.0003	0.8078	-2,799.70	-2,756.55	0.3407	0.0118
Absolute shocks	3,514	0.8084	0.0005	0.8080	-2,803.52	-2,760.36	0.3549	0.0104

Note: ΔR^2 is measured relative to the lagged-volatility control model. Lower AIC and BIC values indicate a more favourable balance between model fit and parsimony. LB(k) denotes the Ljung-Box residual p -value at lag k . The development sample excludes the untouched test period. All specifications retain residual serial correlation at lag 20; HAC standard errors are therefore used for inference.

Lagged log conditional volatility is strongly significant in both shock specifications. Its estimated coefficient is approximately 0.90, indicating that the selected volatility path is dominated by persistent internal dynamics. Adding financial shocks increases R^2 only marginally. Relative to the lagged-volatility control model, the incremental increase is 0.0003 for the signed-shock specification and 0.0005 for the absolute-shock specification.

The information criteria reinforce this cautious interpretation. Although the absolute-shock model produces the highest R^2 , the control-only specification retains the lowest AIC and BIC values. The additional shock regressors therefore provide limited incremental explanatory content after volatility persistence is controlled for.

Most individual shock coefficients are not statistically significant at the 5 and are weakly significant at the 10 and a p -value of 0.0735. In the absolute-shock specification, the magnitude of lagged VN-Index returns has a positive coefficient of 0.0106 and is statistically significant at the 5 movements are therefore associated with higher subsequent USD/VND conditional volatility after controlling for the persistence of the volatility process.

The HAC estimates must be interpreted as supplementary reduced-form associations. The dependent variable is a generated conditional-volatility series obtained from a first-stage GJR-GARCH model. Moreover, the residual Ljung–Box p -values remain below 0.05 at lag 20. HAC standard errors reduce the effect of residual heteroskedasticity and autocorrelation on inference, but they do not convert the reported associations into structural causal effects.

3) *Weekly Recursively Identified VAR Results:* A weekly recursively identified VAR analysis is conducted to examine whether financial shocks contain predictive information for USD/VND returns and selected conditional volatility at a lower frequency. The daily shock variables and FX returns are aggregated by weekly summation, while log selected conditional volatility is aggregated by weekly averaging. The systems are estimated on the development period only. The contemporaneous daily shocks are reconstructed from the audited feature-engineering output without reading into the untouched test sample.

All weekly variables are standardized before estimation. The recursive ordering is specified as follows:

$$\Delta VIX \rightarrow \Delta US10Y \rightarrow r^{DXY} \rightarrow \Delta OIL \rightarrow r^{VNI} \rightarrow \text{Target.} \quad (41)$$

Lag orders from one to eight weeks are evaluated. Only stable candidates are eligible for selection. All evaluated candidates satisfy the stability condition. BIC selects a one-week lag for both target equations. The weekly log-volatility level specification is stable, so the permitted first-difference fallback is not required.

TABLE XV
WEEKLY RECURSIVELY IDENTIFIED VAR MODEL SUMMARY

Target	Transformation	Weekly obs.	Vars.	Lag	BIC	Max. $ \lambda $
USD/VND return	Weekly return	729	6	1	-0.2306	0.1394
Selected conditional volatility	Weekly log-volatility	729	6	1	-0.8483	0.6957

Note: The volatility source is the development-sample ARMA(4,5)–GJR-GARCH(1,1)- t model. Weekly USD/VND returns are obtained by summing daily returns, while weekly log conditional volatility is obtained by averaging daily log volatility within each week. Lag orders from one to eight weeks are evaluated. The selected lag minimizes BIC among stable candidates. Max. $|\lambda|$ denotes the largest absolute companion-matrix eigenvalue for the selected VAR; values below one indicate stability. The recursive ordering is reported in Equation (41).

TABLE XVI
WEEKLY VAR GRANGER-CAUSALITY RESULTS

Shock variable	FX-return p -value	Volatility p -value	FX-return predictive content	Volatility predictive content
ΔVIX	0.2482	0.1349	No	No
$\Delta US10Y$	0.0012	0.5120	Yes	No
r^{DXY}	0.0092	0.4251	Yes	No
ΔOIL	0.4830	0.1304	No	No
r^{VNI}	0.0424	0.7177	Yes	No

Note: The table reports shock-specific Granger-causality p -values from one-lag weekly VAR models estimated on the development sample. “Yes” denotes statistical significance at the 5% level. Granger causality indicates incremental predictive content conditional on the remaining variables in the VAR system; it should not be interpreted as structural causality. Bold values indicate statistical significance at the 5% level.

Changes in the U.S. 10-year Treasury yield, DXY returns, and VN-Index returns contain statistically significant predictive information for weekly USD/VND returns at the 5% level. VIX changes and oil-price changes do not provide statistically significant predictive information for weekly USD/VND returns.

None of the five shocks significantly predicts weekly log selected conditional volatility at the 5% level. Once the strong persistence of the conditional-volatility process has been taken into account, the incremental predictive content of financial shocks is limited.

TABLE XVII
WEEKLY RECURSIVE VAR ORTHOGONALIZED IMPULSE-RESPONSE SUMMARY

Shock	FX return ($h = 0$)	FX return ($h = 1$)	FX peak response (week)	Log volatility ($h = 0$)	Log volatility ($h = 1$)	Volatility peak response (week)
ΔVIX	0.0677	0.0512	0.0677 (0)	0.0204	-0.0154	0.0204 (0)
$\Delta US10Y$	0.0375	0.1385	0.1385 (1)	-0.0068	-0.0327	-0.0327 (1)
r^{DXY}	-0.0048	0.1058	0.1058 (1)	-0.0302	-0.0354	-0.0354 (1)
ΔOIL	-0.0359	-0.0309	-0.0359 (0)	-0.0484	-0.0726	-0.0726 (1)
r^{VNI}	-0.1183	-0.0661	-0.1183 (0)	-0.0159	-0.0017	-0.0159 (0)

Note: The table summarizes Cholesky-orthogonalized impulse-response point estimates from the weekly standardized VAR systems. Responses are reported for the contemporaneous period ($h = 0$), the first subsequent week ($h = 1$), and the largest absolute response over horizons $h = 0, \dots, 12$. The week of the peak absolute response is reported in parentheses. Confidence intervals are not available; the estimates are therefore interpreted descriptively rather than as statistically established structural effects.

The orthogonalized impulse-response estimates provide descriptive evidence on the short-run adjustment pattern. For weekly USD/VND returns, the largest responses are concentrated in the contemporaneous period and the first subsequent week. The strongest point responses are associated with VN-Index returns contemporaneously, U.S. Treasury-yield changes after one week, and DXY returns after one week. The point responses decay rapidly beyond the initial horizons.

For weekly log conditional volatility, the point responses are smaller and generally more gradual. The largest absolute response is associated with an oil-price shock, reaching -0.0726 after one week. DXY and U.S. Treasury-yield shocks also produce negative peak responses after one week. These signs and magnitudes should not be interpreted as statistically established transmission effects. The output does not include confidence intervals, and the recursive identification depends on the ordering imposed in Equation (41). The IRFs therefore serve only as supplementary descriptive evidence.

H. Summary of Main Findings

The empirical results produce four main findings.

First, the training-sample conditional-mean analysis selects ARMA(4, 5) as the only specification that satisfies the adjusted residual-whiteness requirement at lags 10, 20, and 30. The residual ARCH effect remains statistically significant, supporting the subsequent use of conditional-volatility models.

Second, GJR-GARCH(1, 1)- t achieves the lowest validation QLIKE and is locked as the primary forecasting model. After reestimation on the combined training and validation sample, it satisfies the standardized residual diagnostics and obtains a

numerically lower test QLIKE than the three simple variance benchmarks.

Third, the locked model does not dominate every reporting-only comparator on the untouched 2024–2025 test sample. EGARCH-type specifications obtain lower test QLIKE values, and the absolute-shock EGARCH-X model ranks first when squared daily returns are used as the robustness proxy. Relative forecast performance is therefore time-varying and proxy-sensitive. These test rankings do not alter the validation-based model lock.

Fourth, global and domestic financial shocks provide limited and specification-dependent supplementary information. The EGARCH-X point estimates suggest that shock magnitude can matter for volatility forecasting. The HAC analysis identifies a positive association between absolute VNINDEX movements and subsequent conditional volatility, but the incremental improvement in model fit is small. The weekly recursive VAR results show that US Treasury yield changes, DXY returns, and VNINDEX returns predict weekly USD/VND returns, while none of the five shocks significantly predicts weekly conditional volatility.

Overall, USD/VND volatility is primarily characterized by strong internal persistence. Global and domestic financial shocks improve the interpretation of exchange-rate dynamics and can enhance forecast performance under selected specifications and proxies, but their incremental contribution is not uniformly stable across evaluation periods.

V. DISCUSSION, LIMITATIONS, AND CONCLUSION

A. Discussion

This study examines whether global and domestic financial shocks improve one-step-ahead forecasts of USD/VND exchange-rate volatility after the persistence of the exchange rate's own conditional-variance process has been modelled. The empirical strategy is deliberately structured as a forecasting exercise rather than an in-sample model-fitting exercise. The model-ready daily sample is divided chronologically into a training period from 6 January 2010 to 31 December 2021, a validation period from 3 January 2022 to 29 December 2023, and an untouched test period from 2 January 2024 to 31 December 2025. The final test sample is excluded from conditional-mean selection, volatility-model selection, EGARCH-X interpretation, HAC regressions, and weekly recursive VAR estimation.

The preliminary diagnostics justify the use of conditional-volatility models. USD/VND returns are stationary after transformation but exhibit serial dependence, conditional heteroskedasticity, volatility clustering, and pronounced non-normality. These properties are consistent with a return process in which periods of limited exchange-rate movement are interrupted by relatively infrequent but substantial adjustments. Such behaviour is plausible in a managed exchange-rate environment, where market pressure may be partially smoothed during normal periods but may still generate discrete movements during periods of stress.

The conditional-mean analysis selects ARMA(4, 5) using the training sample only. This specification is retained because it is the only estimated candidate that satisfies the predefined adjusted Ljung–Box residual-whiteness requirement at lags 10, 20, and 30. The selection rule does not treat conditional heteroskedasticity as a reason to reject the mean equation. The remaining ARCH effects in the ARMA innovations instead provide the basis for estimating the subsequent volatility models.

The validation-period results identify GJR-GARCH(1, 1)- t as the locked forecasting specification. Its validation QLIKE is -2.6783 , the lowest value among the predefined Student- t candidates and variance benchmarks. The absolute-shock EGARCH-X model ranks second with a QLIKE of -2.6684 , followed by the signed-shock EGARCH-X model with a QLIKE of -2.6046 . The validation ranking therefore supports the GJR-GARCH specification as the primary forecasting model, while also indicating that shock magnitudes can provide useful supplementary information.

After the model-selection decision is completed, the required models are re-estimated using the combined training and validation sample. The locked ARMA(4, 5)–GJR-GARCH(1, 1)- t model remains diagnostically adequate after this development-sample refit. The standardized-residual Ljung–Box p -values are 0.2909 and 0.6156 at lags 10 and 20. The corresponding squared-standardized-residual p -values are 0.0904 and 0.6572, while the ARCH–LM p -value is 0.0878. These results indicate that the locked model captures the main conditional-heteroskedasticity dynamics at the conventional 5

The untouched-test results should be interpreted separately from the validation-based selection decision. The locked GJR-GARCH(1, 1)- t model obtains a test QLIKE of -2.7751 , which is lower than the values produced by the EWMA, 22-day rolling-variance, and historical-variance benchmarks. However, the standard EGARCH(1, 1)- t , custom no-X EGARCH, and both EGARCH-X specifications produce numerically lower test QLIKE values. The standard EGARCH(1, 1)- t model ranks first with a test QLIKE of -2.8511 . This does not justify replacing the locked model after inspecting the test sample. Such replacement would convert the untouched test period into a second validation period and invalidate the forecasting design. The correct interpretation is that relative model performance changes across market regimes and evaluation windows.

The robustness analysis reinforces this conclusion. When squared daily returns are used as the alternative volatility proxy, the absolute-shock EGARCH-X specification ranks first on the untouched test sample with a QLIKE of -2.8973 . The sensitivity analysis across innovation distributions also shows that GJR-GARCH dynamics remain competitive. The Gaussian GJR-GARCH(1, 1) variant obtains the lowest validation QLIKE of -2.6984 , while the Student- t and skewed Student- t variants follow closely. The point rankings are therefore sensitive to the assumed innovation distribution and the selected volatility proxy. The evidence supports a disciplined validation-based model lock rather than ex-post selection of a single universally dominant specification.

The custom EGARCH-X estimates provide supplementary evidence on shock transmission. Both the signed-shock and absolute-shock specifications imply persistent conditional-volatility dynamics, with estimated persistence coefficients of 0.9763 and 0.9566, respectively. The absolute-shock model assigns positive coefficients to the magnitudes of DXY, oil-price, U.S. Treasury-yield, and VN-Index movements, while the coefficient on absolute VIX changes is negative. These point estimates suggest that the informational content of financial shocks depends on their transformation and on the model specification. They should not be interpreted as statistically established causal effects because the exported EGARCH-X coefficient table reports point estimates rather than formal inference.

The dynamic HAC regressions provide a more cautious assessment of shock-volatility relationships. Lagged log conditional volatility has a coefficient close to 0.90 and is highly significant in both the signed-shock and absolute-shock specifications. Internal volatility persistence therefore explains most of the variation in the selected conditional-volatility path. Relative to the lagged-volatility control model, the incremental increase in R^2 is only 0.0003 for the signed-shock model and 0.0005 for the absolute-shock model. Among the shock regressors, only the absolute magnitude of lagged VN-Index returns is statistically significant at the 5

The weekly recursively identified VAR analysis produces a similar conclusion. Stable one-lag systems are selected by BIC for both weekly USD/VND returns and weekly log conditional volatility. U.S. 10-year Treasury-yield changes, DXY returns, and VN-Index returns exhibit statistically significant predictive content for weekly USD/VND returns, with p-values of 0.0012, 0.0092, and 0.0424, respectively. VIX changes and oil-price changes do not significantly predict weekly returns. None of the five financial shocks significantly predicts weekly log conditional volatility at the 5

Taken together, the results support a selective rather than uniform shock-transmission interpretation. USD/VND volatility is primarily characterized by strong internal persistence. Global and domestic financial shocks can improve forecast performance under selected specifications and volatility proxies, but their contribution is not stable across evaluation periods. The most consistent supplementary evidence concerns the relationship between domestic equity-market turbulence and subsequent exchange-rate volatility, together with the predictive relevance of U.S. interest-rate conditions, broad dollar movements, and VN-Index returns for weekly USD/VND returns.

B. Limitations

This study has several limitations. First, the modelling pipeline uses a two-step procedure. The ARMA(4, 5) conditional-mean model is estimated first, and its innovations are subsequently passed into the volatility models. This approach is transparent and facilitates a strict recursive forecasting design, but it does not estimate the conditional mean and conditional variance jointly. A fully joint ARMA-GARCH maximum-likelihood specification could account more directly

for parameter interactions between the mean and variance equations.

Second, the selected ARMA(4, 5) model is relatively high-order for daily exchange-rate returns. It is retained because it is the only candidate that passes the predefined adjusted residual-whiteness gate at lags 10, 20, and 30. Nevertheless, a high-order mean specification may absorb short-run noise and may affect the innovations supplied to the volatility models. Further robustness analysis could compare the forecast rankings obtained under simpler mean filters, including lower-order ARMA specifications.

Third, conditional variance is latent and cannot be observed directly. The primary forecast evaluation therefore uses squared recursive ARMA innovations as the variance proxy, while squared raw returns are used as a robustness proxy. The change in model rankings across these proxies shows that the forecast comparison is proxy-sensitive. Higher-frequency exchange-rate data would allow the construction of realized-volatility measures and could provide a more informative evaluation target.

Fourth, the relative performance of the candidate volatility models varies across evaluation periods. GJR-GARCH(1, 1)- t ranks first during validation but is overtaken by EGARCH-type models during the untouched test period. This variation is economically plausible, but it also indicates model uncertainty. The current analysis reports average loss rankings. Pairwise Diebold-Mariano-West comparisons or a model-confidence-set procedure would be useful extensions for assessing whether the numerical loss differences are statistically distinguishable.

Fifth, the EGARCH-X coefficient estimates are interpreted descriptively. The custom EGARCH-X implementation is designed primarily for directly comparable recursive forecasts and exports coefficient estimates without a complete inferential covariance matrix. Consequently, the coefficient patterns cannot be used to establish statistically significant transmission channels. Future work could add numerical Hessian-based standard errors, bootstrap inference, and alternative shock transformations.

Sixth, the HAC regressions use a generated dependent variable: the log conditional volatility extracted from the locked ARMA(4, 5)-GJR-GARCH(1, 1)- t model. The regressions therefore examine associations with a model-implied volatility path rather than with an observed volatility measure. In addition, the HAC regression residuals retain serial correlation at lag 20. Newey-West standard errors reduce the effect of heteroskedasticity and autocorrelation on inference, but they do not convert the coefficients into structural causal effects.

Seventh, the weekly recursive VAR analysis depends on temporal aggregation and recursive identification assumptions. Weekly aggregation can suppress daily transmission patterns, while the Cholesky ordering imposes a particular contemporaneous structure. The impulse-response functions are reported without confidence intervals. Alternative orderings, bootstrap confidence bands, forecast-error variance decompositions, and additional robustness checks would strengthen the supplementary dynamic analysis.

Finally, the set of domestic financial variables is limited. VN-Index returns capture domestic equity-market conditions but do not fully represent Vietnam's macro-financial environment. Reliable higher-frequency measures of domestic interest rates, foreign-exchange intervention, foreign reserves, capital flows, inflation expectations, and sovereign risk could improve the analysis if consistent data become available.

C. Practical Implications

The findings have several practical implications for short-horizon exchange-rate risk monitoring. First, the locked ARMA(4, 5)–GJR-GARCH(1, 1)- t specification provides a parsimonious benchmark for updating one-step-ahead USD/VND conditional-volatility forecasts. For banks, importers, exporters, and firms with U.S.-dollar-denominated exposures, these forecasts can be incorporated as inputs into daily risk dashboards, exposure-limit reviews, hedge-timing decisions, and stress-scenario calibration. The model should not be interpreted as a complete risk-management system because the present study does not estimate hedge ratios or backtest Value-at-Risk measures.

Second, the variation in model rankings across validation and test periods has an operational implication. Shock-augmented EGARCH-X models should be treated as challenger specifications rather than as automatic replacements for the locked GJR-GARCH benchmark. Their performance should be monitored using rolling validation windows and predefined forecast-loss criteria. A model-governance framework based on periodic re-evaluation is more defensible than selecting a new specification after observing favourable performance in a single evaluation window. This interpretation is consistent with the broader volatility-forecasting literature, which does not establish the universal superiority of more complex models over parsimonious GARCH benchmarks for exchange-rate data [6].

Third, the supplementary shock analysis can support the design of an early-warning dashboard. Large absolute VN-Index movements may be monitored as a domestic indicator of elevated next-day USD/VND volatility. Changes in U.S. Treasury yields and DXY returns may also be included as monitoring variables for weekly USD/VND return risk. These variables should be treated as predictive warning indicators rather than as structural causal drivers or direct policy instruments. The incremental explanatory contribution of the shock variables remains limited after controlling for volatility persistence, and the weekly impulse-response estimates are descriptive rather than inferential.

Finally, the policy implication is deliberately limited. The persistence of USD/VND volatility suggests that periods of financial stress may continue to affect short-horizon risk conditions after the initial shock has weakened. Model-based volatility forecasts can therefore complement broader macro-financial surveillance based on market liquidity, capital flows, foreign reserves, and intervention data. The results do not identify an optimal exchange-rate band, intervention threshold, or monetary-policy response. They instead support the use of

disciplined, regularly updated volatility monitoring as one input into policy assessment.

D. Conclusion

This study evaluates USD/VND exchange-rate volatility forecasts using a strict chronological train–validation–test design. The results confirm that USD/VND returns exhibit the statistical properties that motivate conditional-volatility modelling, including serial dependence, conditional heteroskedasticity, volatility clustering, and fat-tailed innovations. The ARMA(4, 5) conditional-mean specification is selected on the training sample and used to generate recursive innovations for the volatility-forecasting exercise.

Among the predefined Student- t forecasting candidates, GJR-GARCH(1, 1)- t achieves the lowest validation QLIKE and is locked before the untouched test period is evaluated. After development-sample refitting, the locked model remains diagnostically adequate and produces lower test QLIKE than the simple variance benchmarks. Reporting-only EGARCH-type comparators obtain lower test losses, but these ex-post rankings do not alter the validation-based model lock. The result indicates that volatility-model performance is time-varying rather than universally dominated by a single specification.

The shock-augmented results are informative but limited. The absolute-shock EGARCH-X model performs strongly under selected evaluation settings, ranking second during validation and first in the squared-return robustness exercise. However, the HAC regressions show that internal volatility persistence explains most of the selected volatility path. The clearest supplementary association is the positive relationship between absolute VN-Index movements and subsequent USD/VND conditional volatility. Weekly recursively identified VAR results show predictive relevance for U.S. Treasury-yield changes, DXY returns, and VN-Index returns in the weekly USD/VND return equation, while none of the examined shocks significantly predicts weekly conditional volatility.

Overall, USD/VND volatility is primarily driven by persistent internal dynamics. Global and domestic financial shocks provide incremental information in selected specifications, but their contribution is modest, proxy-sensitive, and not uniformly stable across periods. The main practical implication is that disciplined out-of-sample validation is essential: richer shock-augmented models can be useful, but they should not replace simpler locked specifications solely because they perform better in a single test window.

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